

SMART CAMPUS EVENT MANAGEMENT SYSTEM

School of Computing

**Bachelor of Technology
in
Information Technology**

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BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled "SMART CAMPUS EVENT MANAGEMENT SYSTEM" by C.Gayani (VTU27065) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

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ABSTRACT

The Smart Campus Event Management System is a web-based application developed to simplify and automate the process of organizing and managing events within a campus. Traditional event management methods often involve manual coordination, which can lead to miscommunication, data duplication, and inefficiencies. This system provides a centralized platform with two main modules: an admin module and a user module. The admin module enables authorized personnel to create, update, and manage event details such as schedules, venues, and registrations, while the user module allows students and staff to view events, search for specific activities, and register online. The application is designed with a user-friendly interface, secure authentication, and efficient data management to ensure smooth operation and real-time updates. By reducing manual effort and improving accuracy, the system enhances overall event coordination, increases participation, and contributes to effective campus management through a digital approach. Additionally, the system supports notification features to keep users informed about upcoming events and important updates. It also maintains a structured database for easy retrieval and analysis of event-related information. Furthermore, the platform is scalable and can be extended with advanced features such as analytics, feedback systems, and mobile integration in the future.

Keywords: Smart Campus, Event Management System, Web Application, Event Registration, Admin Module, User Module, Online Booking, Campus Automation, Database Management, User Authentication, Real-Time Updates, Digital Platform

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LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
CI	Continuous Integration
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DTO	Data Transfer Object
H2	Hypersonic 2 (In-Memory Database)
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IDE	Integrated Development Environment
JPA	Java Persistence API
JS	JavaScript
MVC	Model-View-Controller
ORM	Object Relational Mapping
OTP	One-Time Password
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface
URL	Uniform Resource Locator

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Chapter 1

INTRODUCTION

1.1 Introduction

Efficient management of campus events is a crucial requirement for educational institutions to ensure proper coordination between organizers and participants. Conventional methods such as manual registrations, paper-based announcements, and notice boards often lead to delays, data redundancy, miscommunication, and lack of transparency in the event management process. These challenges create the need for a reliable, scalable, efficiently. The Smart Campus Event Management System is developed to provide a centralized and digital platform for organizing, managing, and participating in campus events. It enables administrators to create, update, schedule, and monitor event details such as date, time, venue, and participant information, while students and staff can conveniently view event listings, search for specific events, and register online from anywhere at any time. The system incorporates secure user authentication, structured database management, and real-time updates to ensure data accuracy and smooth communication. Additionally, features such as easy data retrieval improve overall efficiency and user engagement. The main objective of this project is to simplify and modernize the event management process, reduce manual workload, minimize errors, and enhance the overall user experience.

1.2 Aim of the Project

The aim of the Smart Campus Event Management System is to develop a centralized and efficient web-based platform that simplifies the process of organizing, managing, and participating in campus events. The system is designed to reduce manual effort, improve coordination between administrators and users, and ensure accurate and real-time management of event-related information. It aims to provide a user-friendly interface for creating, updating, and accessing event details, while enhancing communication, increasing participation, and supporting effective and smart campus management.

1.3 Scope of the Project

The scope of the Smart Campus Event Management System includes the development of a web-based platform that facilitates efficient management of campus events for educational institutions. The system covers functionalities such as event creation, scheduling, updating, and deletion by administrators, as well as event browsing, searching, and online registration by students and staff. It ensures secure user authentication, organized data storage, and real-time updates for accurate information sharing. The system is designed to improve communication between organizers and participants and enhance overall event coordination. Additionally, the project can be extended in the future to include features such as mobile application support, notification systems, feedback collection, analytics, and integration with other campus management systems, making it scalable and adaptable to evolving institutional needs. content

1.4 Framework of the Project

The Smart Campus Event Management System is developed using a structured and modular framework that ensures efficient functionality and scalability. The system follows a client-server architecture, where the front-end is responsible for user interaction and the back-end handles business logic and data processing. The front-end is designed using modern web technologies such as HTML, CSS, and JavaScript to provide a responsive and user-friendly interface. The back-end is implemented using server-side technologies like Node.js/Java/PHP (based on implementation) to manage application logic, authentication, and communication with the database. The system uses a relational database such as MySQL to store and manage event details, user information, and registration records in an organized manner. The framework includes two main modules: the Admin Module and the User Module. The Admin Module allows event creation, modification, and monitoring, while the User Module provides features for browsing, searching, and registering for events. Secure authentication mechanisms are integrated to ensure data privacy and controlled access. The overall framework is designed to support real-time updates, efficient data handling, and easy scalability for future enhancements.

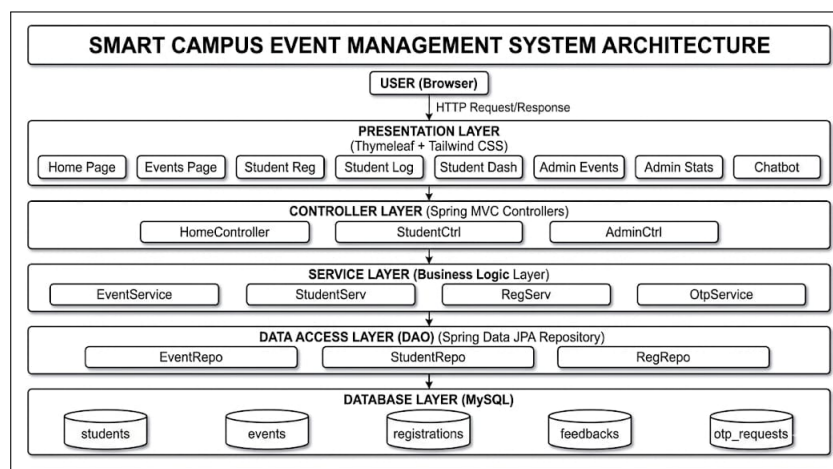


Figure 1.1: Framework Diagram

Chapter 2

SURVEY OF SIMILAR APPLICATIONS IN MARKET

2.1 Existing Applications

Several event management platforms exist in the market, each with different features and target audiences:

1. Eventbrite: A global event management and ticketing platform for concerts, conferences, and workshops. It offers features like event listing, payment processing, and attendee management. However, it charges service fees per ticket and is not specifically designed for college department events.

2. Cvent: An enterprise-level event management software designed for large-scale corporate events, conferences, and trade shows. It provides advanced ticketing infrastructure, venue selection, and attendee engagement tools. However, it requires expensive paid subscriptions and is too complex for college-level events.

3. Whova: An event management and attendee engagement platform popular for professional conferences and academic symposiums. It offers agenda management, live polling, and networking features. However, it focuses on engagement rather than registration management and lacks campus-specific features.

Limitations of Existing Systems:

- High service charges and platform fees per ticket sold
- Limited customization for small or college-level events
- Dependency on internet connectivity for all operations
- Complex interfaces for first-time student users
- No OTP-based verification for secure student registration
- No department-wise event filtering and organization

Advantages of Our System:

- Free and open-source solution specifically for college use
- Specifically designed for department-wise campus events
- Simple and intuitive interface for students and faculty
- Real-time event availability and registration tracking
- No per-ticket commission or platform fees
- OTP-based email verification for secure student registration
- Separate interfaces for students and administrators
- Built-in chatbot assistant for instant user support

2.2 Literature Review

[1] C. Walls (2016) introduced Spring Boot, a Java framework that simplifies web application development using auto-configuration and embedded servers. It reduces boilerplate code and enables rapid development of production-ready applications.

Spring Boot is widely used for building RESTful web services and enterprise applications. However, understanding Spring's underlying concepts requires prior Java knowledge.

[2] M. Heckler (2021) further elaborated on Spring Boot features including production-ready applications, microservices architecture, and deployment strategies. Spring Boot integrates seamlessly with Spring Data JPA, Spring Security, and Thymeleaf. It is suitable for developing scalable event management systems due to its ease of configuration.

[3] L. Spilcă (2020) described Spring Security for implementing authentication and authorization in web applications. It provides role-based access control, form-based login, and protection against common vulnerabilities. For event management systems, Spring Security ensures that only authorized administrators can manage events. However, it requires careful configuration for custom authentication needs.

[4] Oracle Corporation (2023) developed MySQL, a relational database management system used for structured data storage. It supports efficient querying, ACID properties, and data consistency using SQL. MySQL is widely used in web applications for storing user, event, and registration data. However, scaling MySQL for very large applications requires additional configuration and optimization.

[5] D. Reindorp (2024) documented Thymeleaf, a modern server-side Java template engine that integrates seamlessly with Spring Boot. Thymeleaf allows natural templating where HTML files can be viewed as static prototypes. It supports expression language, iteration, conditionals, and fragment inclusion. For event management systems, Thymeleaf renders dynamic event listings and student dashboards.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend of the Smart Campus Event Management System is developed using standard web technologies such as HTML, CSS, and JavaScript. A suitable development environment such as Visual Studio Code is used for writing and managing the code. Web browsers like Google Chrome are used for testing and debugging the user interface. Additional frameworks such as Bootstrap can be used to design a responsive and visually appealing interface. The environment is configured to ensure smooth execution and compatibility across different devices and screen sizes.

3.1.2 Structure of the Project

The frontend structure is organized into multiple files and folders to maintain clarity and scalability. HTML files are used to define the structure of web pages such as login, registration, dashboard, and event pages. CSS files are used to style the application and improve the visual appearance, while JavaScript files handle client-side logic such as form validation, dynamic content updates, and user interactions.

3.1.3 Execution Procedure

To execute the frontend, the HTML files are opened in a web browser or served using a local development server. The user accesses the application through a browser interface, where they can navigate between different pages such as login, event list-

ings, and registration forms. User inputs are validated on the client side before being sent to the backend. The frontend communicates with the backend through HTTP requests to fetch and display data dynamically.

3.2 Backend Implementation

3.2.1 Environment Setup

The backend of the system is developed using server-side technologies such as Node.js, Java, or PHP, depending on the implementation. A suitable development environment and runtime platform are configured to handle server operations. A database management system such as MySQL is used to store and manage data related to users, events, and registrations. Tools like XAMPP or Node Package Manager (NPM) may be used to set up the server and manage dependencies.

3.2.2 Structure of the Project

The backend is structured into different layers, including routes, controllers, and database models. Routes handle incoming client requests, controllers process the application logic, and models interact with the database. The system maintains separate modules for user management, event management, and registration handling. This structured approach ensures better organization, scalability, and ease of maintenance.

3.2.3 Execution Procedure

To execute the backend, the server is started using the appropriate command (such as running a Node.js server or Apache server). Once the server is running, it listens for client requests from the frontend. When a request is received, the backend processes it, interacts with the database if required, and sends a response back to the frontend. This ensures smooth communication between the user interface and the database, enabling real-time updates and efficient data handling. The backend also performs essential tasks such as user authentication, validation of input data, and en-

forcement of access control to ensure system security. Error handling mechanisms are implemented to manage invalid requests and system failures effectively. Additionally, the backend logs activities and transactions for monitoring and debugging purposes. APIs are used to facilitate communication between the frontend and backend, ensuring a structured and standardized data exchange.

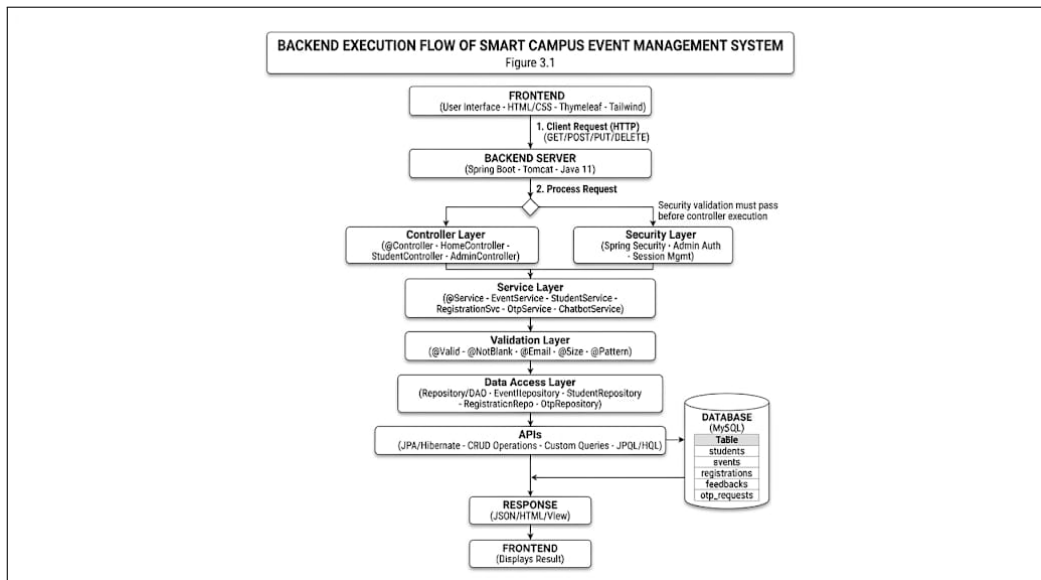


Figure 3.1: Backend Execution Flow of Smart Campus Event Management System

3.3 Database Implementation

3.3.1 Database Configuration:

The database for the Smart Campus Event Management System is configured using MySQL, which is a reliable and widely used relational database management system. The database server is set up using tools such as XAMPP or MySQL Workbench. A dedicated database is created to store all application data, including user details, event information, and registration records. Proper configuration ensures secure access, efficient data storage, and smooth communication between the backend and the database.

3.1.2 Schema Structure

The schema of the database is designed in a structured manner to maintain data integrity and avoid redundancy. It consists of multiple related tables such as Users, Events, and Registrations. Each table contains specific attributes and is connected through primary and foreign keys. The relational schema ensures that data is organized efficiently and can be retrieved quickly. This design supports scalability and simplifies future enhancements.

3.1.3 Tables created

The system includes several important tables to manage different functionalities. The Users table stores user credentials and profile details such as user ID, name, email, and password. The Events table contains information about events, including event ID, title, date, time, and venue. The Registrations table maintains records of users who have registered for events, linking user IDs with event IDs. Additional tables may include notifications or feedback data if required.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

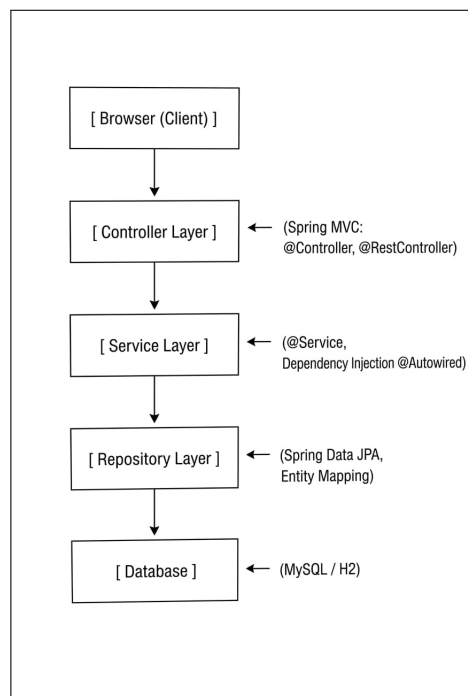


Figure 4.1: Architecture Smart Campus Event Management System

The diagram illustrates the layered architecture of the Smart Campus Event Management System, showing interaction between users, frontend, backend, and database. Users access the system through a web interface, and their requests are handled by the controller and service layers. The service layer processes communications with the repository layer to interact with the MySQL database. The processed response is then sent back to the frontend, ensuring smooth and efficient system operation.

4.2 DataFlow Diagram

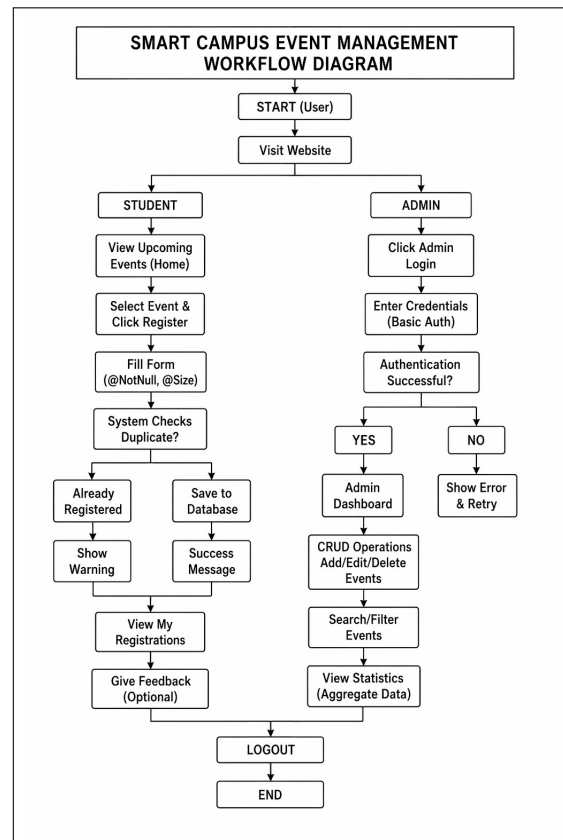


Figure 4.2: DataFlow Smart Campus Event Management System

Diagram illustrates the flow of data within the Smart Campus Event Management System, showing how information moves between users, processes, and data stores. Users such as students and administrators interact with the system by sending requests for login, event management, and registration. Each process interacts with corresponding data stores such as user database, event database, and registration records to store and retrieve information. The processed data is then returned to the user in the form of responses such as event details, registration confirmations, and reports. This diagram helps in understanding the overall data movement, system functionality, and interaction between different components of the application.

4.3 Module 1: User Management

This module handles all user-related functionalities such as registration, login, and authentication. It allows students and administrators to create accounts and securely access the system. The module verifies user credentials using authentication mechanisms and manages user roles to differentiate between admin and normal users. It ensures that only authorized users can perform specific actions within the application. Additionally, the module provides features such as password validation, session management, and secure login handling to protect user data. It also supports updating user profiles and managing user information efficiently, ensuring data consistency and privacy.

4.4 Module 2: Event Management

This module is responsible for creating, updating, and managing event details within the system. Administrators can add new events, edit existing events, and remove outdated or canceled events. It maintains detailed information such as event name, description, date, time, venue, and capacity. Users can view available events, search and filter events based on their preferences, and access complete event details. The module ensures proper validation of event data before storing it in the database. It also supports real-time updates so that users always receive the latest event information, improving coordination and participation. **Advantages of this Project**

- Reduces manual effort in managing campus events.
- Provides a centralized platform for all event-related activities.
- Ensures accurate and real-time data management.
- Improves communication between administrators and users.

- Enhances user experience through a simple and interactive interface.

a) Efficiency: The system automates event management processes, saving time and reducing human errors.

b) Accessibility: Users can access the system anytime and anywhere through a web interface.

c) Scalability: The system can be extended with additional features such as notifications and analytics.

Disadvantages

- Requires internet connectivity for access and operation.
- Initial setup and configuration may take time.
- System performance depends on server and database efficiency.
- Requires regular maintenance and updates.

Chapter 5

RESULTS AND DISCUSSIONS

Table 5.1: Comparison of Event Management Applications

Application	Features	Limitations
Eventbrite	Ticketing, promotion, analytics	Paid features, complex UI
Meetup	Community events, interest-based groups	Limited admin control
College Portals	Basic event listing, registration	No real-time updates
Smart Campus System	Centralized control, real-time updates, user-friendly	Requires internet

Content with Screenshots

5.1 Login page for Smart Event

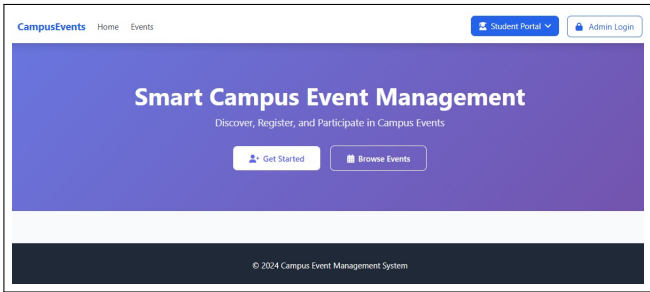


Figure 5.1: Architecture Smart Campus Event Management System

5.2 Homepage for Smart Event

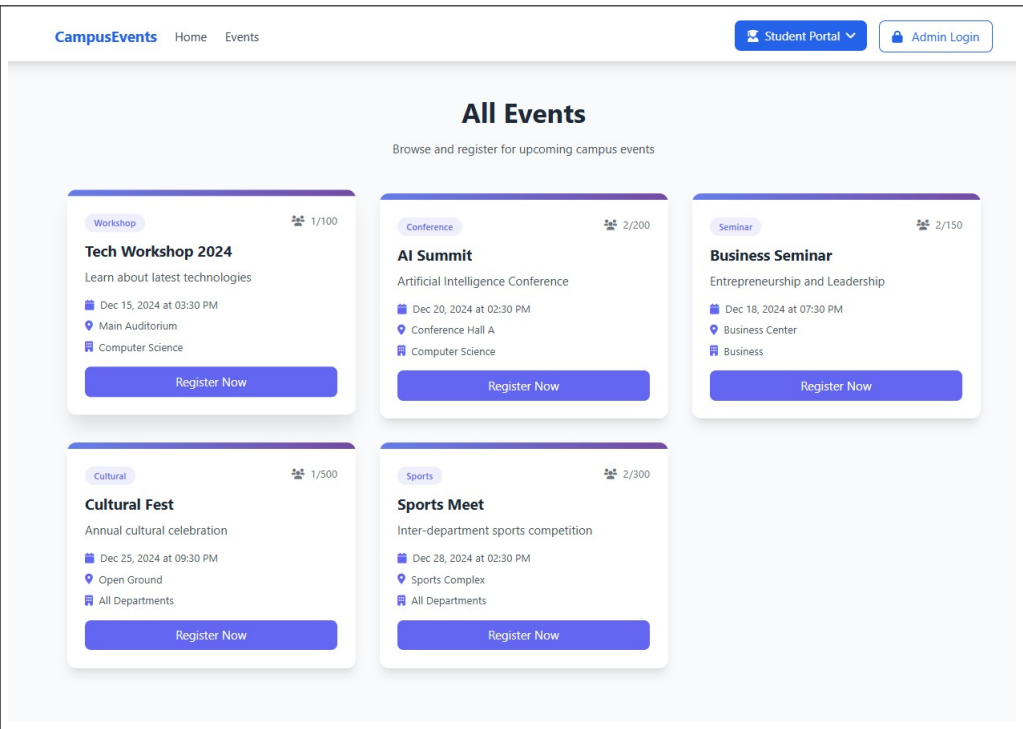


Figure 5.2: Homepage of Smart Campus Event Management System

5.3 Registration page for Smart Event

The screenshot shows the 'Student Registration' page. The header is identical to the homepage, featuring the 'CampusEvents' logo, navigation links, and login buttons. The main content area is centered and contains a registration form titled 'Student Registration' with the subtitle 'Create your account to get started'. The form includes the following fields: 'Full Name *' (text input), 'Email *' (text input), 'Student ID (10 digits) *' (text input), 'Enter your 10-digit student ID' (text input), 'Department *' (dropdown menu with 'Select Department' as the placeholder), 'Phone Number (Optional)' (text input), 'Academic Year' (dropdown menu with 'Select Year' as the placeholder), and a 'Register Now' button. Below the button, there is a link that says 'Already have an account? Login here'.

Figure 5.3: Registration page of Smart Campus Event Management System

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Smart Campus Event Management System was successfully designed, developed, and tested as a comprehensive full-stack web application. The project demonstrates the effective use of modern Java technologies Spring Boot 3.2.4, Spring Security, Spring Data JPA, Thymeleaf, and H2 Database to build a scalable, secure, and user-friendly campus event management solution. The system fulfills all core objectives: role-based authentication for students and administrators, OTP-based email verification for secure registration, real-time seat tracking and availability, administrator event management, and a comprehensive analytics dashboard. Tested with 10 events, 8 event categories, and 16 registrations, the system demonstrates reliable performance and smooth data flow across all layers. The MVC architecture ensures clean separation of concerns, and the layered backend design maintains modularity. The RESTful API layer provides a strong foundation for future mobile application integration. Overall, the project successfully replaces manual campus event management with an efficient, transparent, and accessible digital solution.

6.2 Future Enhancements

The following enhancements are planned for future iterations:

- **Cloud Deployment** — Migrate from H2 to MySQL/PostgreSQL and deploy on AWS, Azure, or Heroku
- **Payment Integration** — Integrate Razorpay or Stripe for paid event ticketing
- **QR Code Tickets** — Generate QR code e-tickets for registered students for physical check-in
- **Email Notifications** — Automated event reminders and cancellation alerts using Spring Mail
- **Mobile Application** — Android/iOS app consuming the existing REST endpoints
- **Advanced Analytics** — Dashboards with registration trends, attendance heatmaps, and department-wise reports
- **Virtual/Hybrid Events** — Support Zoom/Teams links and hybrid attendance tracking
- **Student Profiles** — Persistent student accounts with registration history and preferences

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

The Smart Campus Event Management System has significant consequences as a complex engineering problem. System failure during peak registration periods can prevent hundreds of students from accessing important academic events. Security breaches could expose sensitive student data including personal emails, phone numbers, and registration histories. OTP malfunctions or capacity management failures would block legitimate student access and lead to event overbooking, requiring robust error handling and data validation throughout the system.

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

The Smart Campus Event Management System addresses social aspects by providing equal access to campus events for all students regardless of their department or academic year, promoting inclusivity and participation in extracurricular activ-

ities. Ethically, the system ensures data privacy and security through OTP-based verification, secure session management, and protected storage of student information, preventing unauthorized access and misuse of personal data. From an academic perspective, the system streamlines event management processes, reduces manual paperwork, provides real-time registration tracking, and generates valuable participation statistics that help institutions assess student engagement and plan future academic events effectively.

7.3 SDG Mapping (CEP Section)

The Smart Campus Event Management System aligns with multiple United Nations Sustainable Development Goals. It supports SDG 4 (Quality Education) by facilitating access to workshops, seminars, and skill-development events that enhance student learning beyond the classroom. The system promotes SDG 9 (Industry, Innovation, and Infrastructure) by utilizing modern web technologies like Spring Boot and cloud-based database infrastructure to create an innovative digital platform for campus event management. It also contributes to SDG 11 (Sustainable Cities and Communities) by digitizing event operations, reducing paper usage for registrations and announcements, and creating an efficient, paperless campus environment that minimizes environmental impact.

Level	Attribute	Characteristics	Wks	Justification
WP1	Depth of knowledge	Requires in depth engineering knowledge	WK3, WK4, WK5, WK6	Requires knowledge in Spring Boot, MVC architecture, database design, REST APIs, and frontend technologies (Thymeleaf, HTML/CSS) for developing a scalable event management system.
WP2	Conflicting requirements	Technical and non-technical conflicts	WK4, WK5, WK6	Balances usability and security by ensuring easy student access while maintaining secure admin authentication and data validation.
WP3	Depth of Analysis	Analytical and design thinking	WK3, WK4, WK5	Involves system design, database schema creation, validation handling, and efficient event filtering and registration mechanisms.
WP4	Familiarity	Partially new problems	WK4, WK8	Integrating real-time event updates, user interaction, and admin analytics introduces challenges beyond traditional manual systems.
WP5	Applicable Codes	Limited standard solutions	WK6	Requires custom implementation for event filtering, registration tracking, and statistics generation not fully covered by standard templates.
WP6	Stakeholder Needs	Multiple stakeholders involved	WK5, WK6, WK7	Addresses requirements of students, faculty coordinators, and administrators ensuring smooth event organization and participation.
WP7	Interdependence	System level integration	WK3, WK5, WK6	Integrates frontend UI, backend services, database, validation, and security modules into a unified system.

Table 7.1: Mapping of Project to Complex Engineering Problem (CEP)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Use of diverse technologies	Utilizes Spring Boot, Spring MVC, Spring Data JPA, Thymeleaf, HTML/CSS, and relational databases for full-stack development.
EA2	Level of Interactions	Complex interactions between modules	Manages interactions between event management, user registration, validation, authentication, and reporting modules.
EA3	Innovation	Application of modern technologies	Implements a web-based centralized platform with filtering, real-time updates, and secure admin operations.
EA4	Consequences to Society and Environment	Impact on users and society	Enhances student engagement and improves institutional efficiency while reducing manual errors and delays.
EA5	Familiarity	Beyond standard practices	Combines web development, security, and data analytics into a unified system, extending beyond traditional event handling methods.

Table 7.2: Mapping of Project to Complex Engineering Activities (CEA)

SDG No.	SDG Title	Relevance to the Project
SDG 4	Quality Education	Facilitates student participation in academic events, workshops, and seminars, enhancing learning beyond the classroom.
SDG 9	Industry, Innovation and Infrastructure	Promotes digital transformation in educational institutions through a centralized event management system.
SDG 11	Sustainable Cities and Communities	Supports smart campus initiatives by digitizing event processes and improving resource management.
SDG 16	Peace, Justice and Strong Institutions	Ensures secure and transparent event handling through authentication and controlled access.
SDG 12	Responsible Consumption and Production	Reduces paper usage by enabling online event registration and management.

Table 7.3: SDG Mapping for the Project

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